

Optimizing Color Preservation in Artistic Style Transfer with VGG-19 and Luminance Transfer Approach

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Abstract—Artistic Style Transfer(AST) a fascinating aspect of Neural Style Transfer(NST) has redefined the creative manipulation of digital images by imbuing photographs with the aesthetic characteristics of famous paintings. However a persistent challenge in NST—color distortion—often compromises the color fidelity of the content image during the style transfer process. This paper introduces an innovative approach employing the VGG-19 neural network to extract style features while preserving the color harmony of the content image through luminance transfer. This method ensures that the original color palette remains unaltered achieving an authentic blend of style and content characteristics. In our experimentation, we applied the innovative artistic style transfer technique to a variety of content images, incorporating the aesthetic elements from renowned paintings. Notably, our method successfully preserved the color fidelity of the content images throughout the style transfer process. This approach contributes significantly to the field by offering a improved technique for NST that harmonizes artistic style with natural coloration serving a valuable asset for artists and technologists.

Index Terms—Computer vision, Neural Style Transfer, Convolution Neural Networks, VGG-19 and Artistic Style Transfer.

I. INTRODUCTION

Neural Style Transfer (NST) [6] marks a revolutionary stride in computer vision, introducing a transformative approach that merges digital imagery with the aesthetic styles [19] found in various artworks. This groundbreaking technique, particularly in the domain of Artistic Style Transfer (AST) [14], has redefined our perception and engagement with digital media. AST stands as more than just a technical accomplishment; it serves as a creative tool, seamlessly fusing historical artistic styles with modern digital formats. This unique blend opens up novel possibilities for artists, designers, and technologists, offering avenues for creative expression and digital experimentation previously unexplored.

In the expansive field of image processing and computer vision, Convolutional Neural Networks (CNNs) [1] play a

pivotal role, with VGG-19 [16] standing out due to its depth and robust pattern recognition capabilities. VGG-19 [16], a 19-layer deep network, is recognized for its proficiency in extracting complex features [2] from images, making it particularly suitable for tasks such as Artistic Style Transfer. Its layered structure enables an intricate analysis of style features [14], rendering it a preferred choice for researchers and artists delving into digital image processing.

Despite the progress made in style transfer using CNNs [7] like VGG-19 [16], a notable challenge [17] persists color distortion. Traditional style transfer methods [6] often compromise the original color palette [3] of the content image, resulting in a loss of authenticity in the final output. To overcome this, luminance transfer [13] techniques have emerged as a solution. These techniques focus on preserving [5] the luminance, brightness and tonal quality of the content image while incorporating style features from the chosen style image. This nuanced approach ensures that the color harmony of the original image [3] is maintained, significantly enhancing the overall quality and authenticity of the style-transferred image.

The motivation driving this research stems from a desire to augment the artistic integrity of digital compositions within Neural Style Transfer. Recognizing the limitations of existing methods [14], particularly in preserving the color harmony [5] of original images, this study seeks to explore and refine the techniques employed in style transfer. By synergizing the strengths of VGG-19 for detailed feature extraction with advanced luminance transfer methods [10], this research aims to address the challenge of color distortion, contributing to the development of more accurate and visually appealing digital artworks. This collaborative effort not only advances technological capabilities in image processing but also enriches the artistic value of digitally transformed images.

In the subsequent sections, we delve into the details of our methodology, the underlying principles of VGG19, luminance

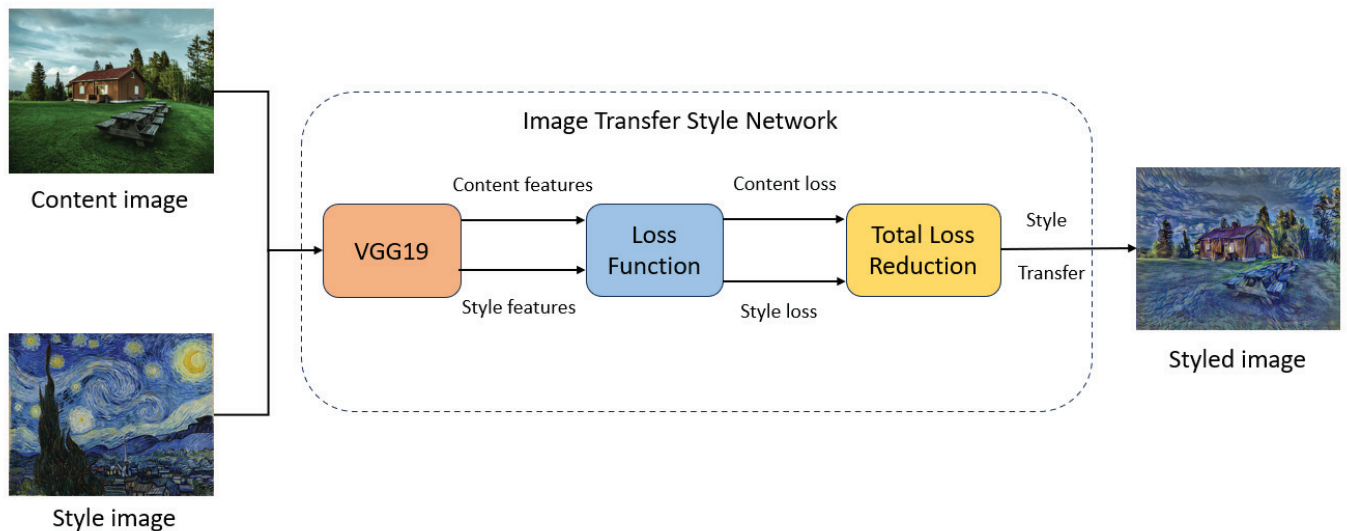


Fig. 1: Neural Style Transfer

transfer, and the intricacies of the stylization process [20]. The paper concludes with a compelling presentation of results, affirming the success of our approach in bridging the gap between style transfer and color fidelity.

II. BACKGROUND

The advent of Neural Style Transfer (NST) [14] has heralded a transformative era in the domain of computer vision, fundamentally altering how we perceive and interact with digital imagery. This groundbreaking technique not only facilitates the seamless fusion of artistic styles with digital images but also lays the foundation for the emergence of Artistic Style Transfer (AST) [6], a specialized branch of NST. AST takes the core principles of NST and refines them, with a specific emphasis on replicating the unique and distinctive styles found in renowned artworks, all within the realm of digital media. At the heart of these pioneering techniques lies the remarkable concept of Convolutional Neural Networks (CNNs) [1], a class of deep learning models inspired by the intricate workings of the human visual system. These neural networks possess a remarkable ability to process and analyze images with exceptional precision. They excel in discerning intricate patterns, extracting fine grained details, and understanding the visual content of images. Among the diverse spectrum of CNN architectures, VGG-19 [16] stands out as a prominent and influential player. This 19-layer deep network has garnered acclaim for its exceptional proficiency in extracting and representing style features from images. Its depth and robustness make it an indispensable tool in the realm of digital image processing.

Despite the remarkable strides made in NST and AST, a persistent challenge has cast a shadow over these transformative techniques: color distortion [12]. Traditional NST methods have often grappled with an unintended consequence the alteration of the original color palette of the

content image during the style transfer process. This unintended color shift has been a source of concern, as it disrupts the intended color harmony and authenticity of the final output.

To tackle this vexing challenge, luminance transfer [10] techniques have emerged as a beacon of innovation. These advanced methods have been meticulously designed to safeguard the intrinsic brightness and tonal qualities, collectively referred to as luminance, of the content image. Simultaneously, these techniques artfully integrate the style features derived from the selected style image. The result is a finely crafted approach that ensures the preservation [5] of the original image's color harmony. By doing so, it elevates the overall quality and authenticity of the style-transferred artwork, breathing new life into the fusion of art and technology. In this ever evolving landscape of computer vision and digital artistry, understanding the roots of NST and AST, the pivotal role played by CNNs like VGG-19 [16], and the innovative luminance transfer [13] techniques is crucial. These foundational insights serve as the building blocks for our exploration of the intricate world of NST [14], as we unravel its origins, evolution, and profound impact on reshaping the relationship between art and technology in the digital age.

A. Convolutional Neural Networks

In the realm of computer vision, we explore the profound importance of Convolutional Neural Networks (CNNs) [1], delving into their pivotal role and substantial impact on the field. These networks, inspired by the human visual system, possess an extraordinary ability to discern intricate patterns and fine details within images, highlighting their fundamental relevance. The architecture of CNNs, a remarkable feat of engineering, comprises various layers, each with a distinct role, including convolutional layers for adaptively filtering input images, pooling layers for reducing data dimensions,

and fully connected layers for pattern recognition. Each layer collaborates seamlessly to extract features [2] and comprehend image content. Furthermore, CNNs extend their significance beyond image analysis, playing a pivotal role in image classification, interpretation, and understanding. This is exemplified by their transformative applications across diverse domains. In the context of Neural Style Transfer (NST), CNNs are instrumental in analyzing image content and style, facilitating the fusion of two distinct images to create captivating artistic transformations.

B. VGG-19

VGG-19 [16] stands as a significant architecture within the deep learning community, particularly celebrated for its deep convolutional neural network design. It has established a reputation for effectively extracting complex style features [2] from images, making it particularly adept for tasks that require a nuanced understanding of image content and style, such as Artistic Style Transfer (AST). The VGG-19 model is characterized by its depth, encompassing 19 layers that are essential for capturing the intricate details [4] necessary for style transfer.

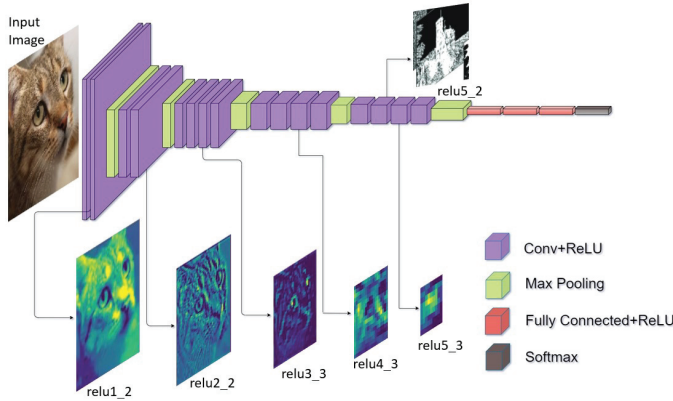


Fig. 2: VGG-19 Architecture for Artistic Style Transfer

As illustrated in Fig. 2, the architecture is composed of multiple convolutional layers [1], each followed by a rectified linear unit (ReLU) to introduce non-linearity, and punctuated by max pooling layers to downsample the image representation, thus reducing the computational load. This cascade of layers is designed to process the visual data progressively, extracting and refining features [2] at every level, from the simplest to the most complex. The final stages of VGG-19 are made up of fully connected layers that consolidate these features into a format suitable for classification tasks, although for AST, the earlier layers are of greater interest as they hold the style information. In practical terms, VGG-19 has been pivotal in the progression of image processing techniques, particularly in the realm of Neural Style Transfer. Its proficiency in decomposing and reassembling the stylistic and content elements of images has revolutionized the creation of digitally altered artwork. VGG-19's influence extends beyond mere image stylization; it also contributes to better understanding

and visualization of complex patterns in educational platforms, enriching the interplay between technology and art.

C. Neural Style Transfer

Neural Style Transfer (NST) [6] is a captivating fusion of technology and artistic creativity, where the unique style of one image is seamlessly applied to another. This intricate process, depicted in Fig. 1, harnesses the power of deep neural networks to craft a distinctive output that preserves [5] the structure of the original image while adopting the visual elements of the chosen artistic style. Pioneered by

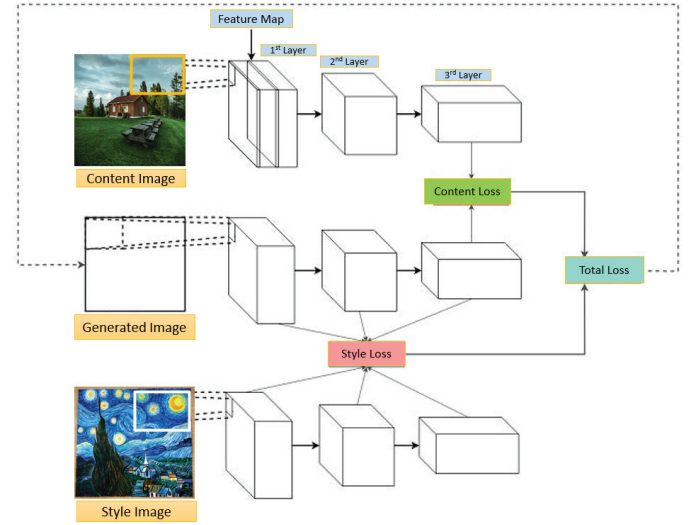


Fig. 3: Style Transferring Process

the author [14], NST employs convolutional neural networks (CNNs) [7] to merge renowned paintings' brushwork and color palettes with everyday photographs. Advancements by researchers have refined spatial application [8] and developed methods to preserve luminance and color characteristics [9] during style transfer. Contributions include histogram losses [20] for stable texture synthesis and global/local optimizations. These studies significantly improved NST quality, enabling faithful reproduction of artistic aesthetics. Fig. 3 shows the style transfer process which involves the feature extraction and loss functions for style and content, guided by the total loss [15], that will result in a beautifully merged image embodying both content and style.

D. Luminance Transfer

Luminance transfer [13] is a pivotal technique in the domain of NST, specifically designed to address the challenge of color distortion that often occurs during the style transfer process. This approach is focused on preserving the original color harmony [3] of the content image while adopting the stylistic features of the style image. It becomes particularly crucial in maintaining the authenticity of the content image's colors, ensuring they remain true to reality after the style transfer.

The implementation of luminance transfer can be illustrated by considering four key images in Fig. 4, the content image (a)



Fig. 4: Luminance Preservation in Style Transfer

depicts a serene house amidst a lush landscape, the style image (b) features the iconic swirling skies of Van Gogh's "Starry Night", (c) shows the result of a direct style transfer onto the content image without luminance preservation, resulting in a dramatic shift in the overall color scheme, and (d) demonstrates the effect of applying luminance transfer, where the style's textural essence is captured while the natural color palette of the content image is retained.

The impact of incorporating luminance transfer is evident when comparing (c) and (d). Without luminance transfer, the colors in the content image can become oversaturated or unnaturally altered, as seen in (c), where the vivid blues and yellows of the style image dominate the scene, overshadowing the natural tones of the content. On the other hand, (d) maintains the color integrity of the content image, such as the warm browns of the house and the greenery of the grass, while still integrating the dynamic brushstrokes and movement of the style image. This nuanced blending enriches the final image with artistic flair without compromising the realistic appearance of the original scene, thereby enhancing the visual experience and broadening the potential applications of NST in various fields, from art curation to therapeutic imagery.

III. METHODOLOGY

The proposed comprehensive approach aims to preserve color harmony [5] during artistic style transfer. Central to this method is the utilization of VGG-19 [16] and the incorporation of luminance transfer [13]. VGG-19 is configured to extract intricate features from both content and style images, and luminance transfer is employed to ensure faithful color reproduction. The synergy of these elements forms a robust framework for achieving nuanced and authentic artistic style transfer.

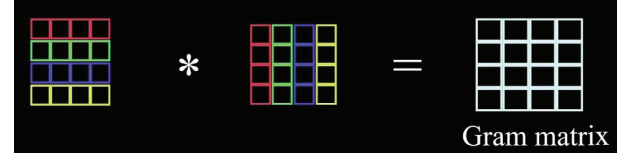


Fig. 5: Gram matrix calculation visual

To implement the VGG-19 model [16], the pre-trained weights are loaded and customized for feature extraction. Specifically, we focus on the convolutional layers [1] to capture detailed information. For style extraction, the relevant layers for harnessing intricate patterns are identified, and for content extraction, the layers that encapsulate meaningful content features are pinpointed. The configuration is visually represented in Fig. 2.

In the context of style extraction, we direct the VGG-19 model to specific convolutional layers, namely relu1-2, relu2-2, relu3-3, relu4-3, and relu5-3. These layers are strategically chosen to capture complex textures and stylistic elements present in the style image [4]. Eq. (1) introduces the Gram matrix [18], a pivotal tool for capturing correlations between different feature maps at a given layer as shown in Fig. 5. Utilizing this Gram matrix, the style loss is calculated as the mean squared difference between the Gram matrices of the style features in the style image and the generated image [15] and is expressed as:

$$\mathcal{L}_s = \frac{1}{4N^2M^2} \sum_{i,j} (G_{ij}^s - A_{ij}^s)^2 \quad (1)$$

Here $\mathcal{L}_s$ is the style loss and (G_{ij}^s, A_{ij}^s) are the Gram matrices of style and generated image. These computations are essential in ensuring that the generated image encapsulates the stylistic patterns inherent in the style image, contributing to the faithful representation of artistic style.

In contrast, content extraction zooms in on a specific convolutional layer, chosen for its unique ability to capture meaningful content features. This deliberate selection allows us to focus on preserving the essence and content of the original image while applying the desired artistic style

$$\mathcal{L}_c = \frac{1}{2} \sum_{i,j} (F_{ij}^c - P_{ij}^c)^2 \quad (2)$$

The content loss ($\mathcal{L}_c$), as defined in Eq. (2), is derived by comparing the features at a specific layer in both the generated image (F_{ij}^c) and the reference content image (P_{ij}^c) [20]. This computation effectively quantifies the disparity between the content features of the generated image and those of the reference content image at the chosen layer. By concentrating on this particular layer, we ensure that the extracted features not only retain intricate patterns from the style image but also preserve the significant content from the reference content image.

The total loss ($\mathcal{L}_{total}$), defined in Eq. 3 encompasses both style and content losses, ensuring a balanced optimization objective. The total loss is expressed as:

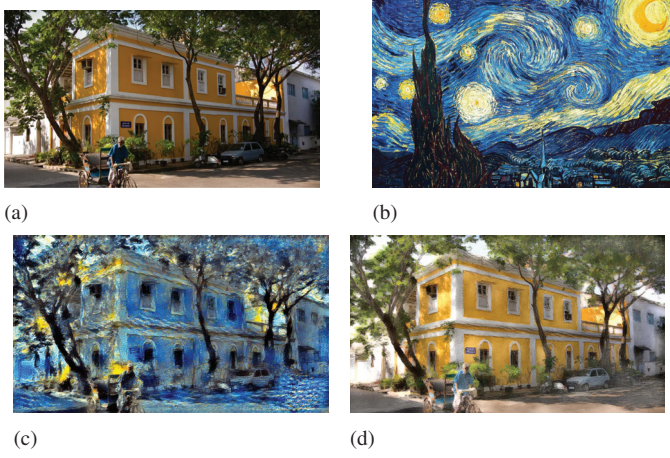


Fig. 6: Luminance Preservation in Style Transfer on 2nd content image with vg-starry-night style image

$$\mathcal{L}_{\text{total}} = \alpha \mathcal{L}_s + \beta \mathcal{L}_c \quad (3)$$

Here, α and β are weights that govern the contributions of style and content to the overall optimization process [11]. This formulation guides the optimization algorithm [14] to generate a stylized image that authentically reflects the style of the style image while preserving essential content characteristics [7].

In addition to the feature extraction process, an innovative approach is introduced to preserve color harmony during the style transfer process. This approach leverages luminance transfer [9] to ensure that the original color palette of the content image remains unaltered. The content image's luminance information is combined with the stylized image's color information, providing a nuanced solution for maintaining color authenticity [19]. The luminance transfer is expressed as:

$$L'_{\text{sty}} = \frac{\sigma_c}{\sigma_s} \cdot (L_{\text{sty}} - \mu_s) + \mu_c \quad (4)$$

Eq. (4) represents luminance transfer in NST, where L'_{sty} is the transformed luminance, σ_c and σ_s are the standard deviations of the content and style luminance, L_{sty} is the original style luminance, μ_s is its mean, and μ_c is the mean luminance of the content image.

The combined methodology, integrating feature extraction with style and content losses and luminance transfer results in a comprehensive approach to artistic style transfer [6]. This approach not only faithfully represents the chosen style and preserves the original color characteristics of the content image but also ensures spatial coherence, producing visually captivating and authentic artistic compositions.

IV. RESULTS

We present a comprehensive exploration of the impact of luminance preservation in Neural Style Transfer (NST) through a visual analysis of Fig. 6 and 7. These collectively offer a vivid comparative study on the effects of luminance preservation [5] in Neural Style Transfer (NST). They are



Fig. 7: Luminance Preservation in Style Transfer on 1st content image with mosaic style image

exemplary visual aids that communicate the nuanced outcomes when NST is applied with and without considerations for color integrity. Fig. 6 focuses on a second content image (Image a), a snapshot of everyday life featuring a building enveloped in the calmness of its natural environment. The colors are lifelike and set the stage for the subsequent style application. The style image (Image b) is Van Gogh's "Starry Night," a masterpiece that is characterized by its energetic swirls and intense color contrasts. In Image (c) of Fig. 6, we witness the outcome of NST when executed without luminance preservation. The result is a visually stimulating transformation of the content image, where the dynamic blues and vivid yellows from "Starry Night" overpower the original scene's color scheme. This approach, while artistically rich, deviates from the natural hues that define the content image's reality. Conversely, Image (d) in Fig. 6 displays the NST with luminance preservation, where the content image's inherent colors are preserved. Here, the original colors [3] of the building and its environment are retained, blending seamlessly with the artistic strokes and shades from "Starry Night." The result is a harmonious merger that maintains the authenticity of the content image while adopting the style image's artistic essence.

Turning to Fig. 7, the narrative unfolds similarly but with different imagery. Image (a) depicts the initial content image, a serene landscape graced with a rustic house and verdant terrain. Image (b) introduces a different style image, a vibrant stained glass design, ready to lend its artistic character to the content image. Image (c) of Fig. 7 shows the direct application of NST [14] without luminance preservation. The content image absorbs the stained glass's vivid patterns and colors, losing some of its original color fidelity [12] in the process. Image (d), however, demonstrates the NST with luminance

preservation. The landscape retains its true colors, even as it takes on the intricate details of the stained glass pattern. The final product is an image that fuses the style's artistic flair with the natural beauty of the content image. When examined side by side, Fig. 6 and 7 underscore the transformative capabilities of NST and highlight the critical role of luminance preservation in safeguarding the color harmony of the content images. They eloquently showcase the technology's ability to enhance digital imagery, affirming that with the right approach, one can create artworks that are both aesthetically pleasing and color-authentic.

V. CONCLUSION

In this research, we have ventured into the intricate domain of style transfer, particularly emphasizing the role of VGG19 [16] in this process. It has presented a comprehensive solution to the common challenge of color distortion in Neural Style Transfer (NST) methods. By integrating luminance transfer [13] with the powerful VGG19 for style extraction, our approach not only achieves captivating artistic style transfer but also meticulously preserves the intricate color palettes inherent to content images. The demonstrated effectiveness of our method in maintaining color harmony signifies a significant stride in refining style transfer techniques. As the field progresses, a promising avenue for further exploration lies in incorporating spatial control [8] mechanisms. Such enhancements could offer more granular control over image transformations, paving the way for richer and more nuanced artistic renditions.

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